



Practical Course - Operating Systems - seL4 & TRENTO

W0 – Organisation

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Agenda

- Content

Content

1. Hardware

1. Computer Architecture (ARM, RISC-V)
2. System on Chip (SoC) -> BCM2837
3. Single Board Computer (SBC) -> RPi 3 B+

2. Bootloader -> U-Boot

3. Kernel: seL4

1. Architecture
2. Components
 1. Capabilities
 2. IPC
 3. ...
3. Basic Driver (Timer, UART, Clock, IRQ, Cache, ...)
4. ...

Content ctd.

4. Building Blocks -> on top of seL4, to be used by CAmkES
 1. Libraries (e.g. libplatsupport, libethdrivers, ...)
 2. Utils
 3. ...
5. Userland: CAmkES -> abstraction from kernel + helper
 1. Architecture
 2. Components
 3. Driver (alignment with RPi 3 B+)
 4. Tutorials

Content ctd.

6. Operating System: TRENTOS-M

1. Build process
2. Architecture
3. OS principles (alignment, e.g. from basic OS courses, book of Tanenbaum)
 1. API
 2. Libraries
 3. Components
4. Core building blocks
 1. Storage
 2. Network
 3. Crypto (Keystore, TLS)
5. How to write your first application

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